Benzene and Aromatic Compounds

Answer key with worked solutions

A. Structure

1. Structure of benzene.

> (a) Kekulé: a hexagon with alternating single and double bonds. Delocalized: a hexagon with a circle inside. (b) Each carbon is **sp^2** hybridized, forming three σ bonds (two C–C and one C–H) in a flat ring, with 120° angles. Each carbon has one unhybridized **p orbital** perpendicular to the ring; the six p orbitals overlap side-on all round, so the six π electrons are **delocalized** above and below the ring.

2. Evidence.

> Any two: **all C–C bonds are the same length** (139 pm, between 154 and 134 pm); the **enthalpy of hydrogenation** is much less exothermic than for three C=C bonds; benzene **does not decolourize bromine water** and reacts by **substitution** rather than addition.

3. Empirical formula.

> In 100 g: C 92.3 g, H 7.7 g.
> $n(\mathrm{C}) = \dfrac{92.3\ \mathrm{g}}{12.01\ \mathrm{g/mol}} = 7.69\ \mathrm{mol}$; $n(\mathrm{H}) = \dfrac{7.7\ \mathrm{g}}{1.008\ \mathrm{g/mol}} = 7.6\ \mathrm{mol}$
> Ratio 7.69 : 7.6 ≈ 1 : 1, so the empirical formula is **CH** (and the molecular formula C_6H_6 = 6 × CH).

B. Stability and reactions

4. Delocalization energy.

> Expected for three C=C: $3 \times (-120\ \mathrm{kJ/mol}) = -360\ \mathrm{kJ/mol}$. Actual: $-208\ \mathrm{kJ/mol}$. Difference: $-208 - (-360) = \mathbf{152\ kJ/mol}$. Benzene releases 152 kJ/mol less energy because it starts **152 kJ/mol lower in energy** (more stable) than a structure with three separate double bonds; the extra stability comes from delocalization.

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5. Reactions.

> (a) **Concentrated HNO_3 + concentrated H_2SO_4**, about **50 °C**: nitrobenzene, $C_6H_5NO_2$ (+ H_2O). (b) **Br_2** with a **$FeBr_3$** (or Fe) catalyst: bromobenzene, C_6H_5Br (+ HBr). Both are **electrophilic substitution**.

6. Names.

> (a) **methylbenzene** (toluene) (b) **benzoic acid** (c) **chlorobenzene** (d) **phenol**

C. Calculations

7. Bromobenzene.

> $n(C_6H_6) = \dfrac{15.6\ \text{g}}{78.11\ \text{g/mol}} = 0.1997\ \text{mol}\ (1:1)$
>
> $m(C_6H_5Br) = 0.1997\ \text{mol} \times 157.01\ \text{g/mol} = \mathbf{31.4\ g}$

D. Challenge

8. Nitration of methylbenzene.

> $n(C_7H_8) = \dfrac{9.21\ \text{g}}{92.13\ \text{g/mol}} = 0.09997\ \text{mol}\ (1:1)$
>
> Theoretical: $0.09997\ \text{mol} \times 137.14\ \text{g/mol} = \mathbf{13.7\ g}$
>
> Percent yield: $\dfrac{10.5\ \text{g}}{13.71\ \text{g}} \times 100\% = \mathbf{76.6\%}$

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